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Chapter 9

The Phonology of Formulaic Sequences: A Review

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It's not what you say, it's the way you say it.

Anonymous

In the past decade, many corpus-based studies have revealed the use of formulaic sequences in spontaneous speech (e.g. Altenberg & Eeg-Olofsson, 1990; Altenberg, 1998; Biber, 2006; Biber & Barbieria, 2007; De Cock, 1998, 2000, 2007). A very popular approach taken by these studies is to examine the most frequent *lexical bundles* (Biber, Johansson, Leech, Conrad, & Finegan, 1999) extracted from a spoken corpus using automatic extraction tools like *WordSmith 4.0* (Scott, 2004). While this approach has revealed many important findings, it has masked the essential properties that distinguish spoken formulaic sequences from written formulaic sequences. In fact, with formulaic sequences in spontaneous speech, a lot of the information crucial for interpreting their meanings is encoded in speech context and prosody. It is quite obvious that it is not only *what* is said that counts but also *how* it is said (see Crystal, 2003). It follows naturally that an investigation of spoken formulaic sequences should go a step beyond listing spoken lexical bundles as if they are merely textual in nature, but should also address the prosodic aspects which give colours to these spoken sequences. This paper presents a summary of the prosodic characteristics of formulaic sequences in speech.

Describing the prosodic patterns of formulaic sequences is a complex task because prosody itself is conditioned by many aspects of context, including neighbouring words, syntactic and lexical structure, semantic focus and emphasis, discourse factors and so on. That is why there was a tendency for prosody research in the past to be done in an experimental setting with participants reading aloud scripted sentences. In this way, the contextual factors mentioned above are easily controlled so that the target

variables can be isolated and examined independently. However, prosody of read aloud texts cannot be treated as a more ‘purified’ version of spontaneous speech with the extraneous variables controlled because, as Chafe (2006) shows, their prosodies are two different varieties. While some researchers may see the contextual variables in spontaneous speech samples as extraneous, they actually reflect the speakers’ real time choices of words and utterances when they are under pressure to process speech online. The way the speakers chunk the utterances, stress individual words, repeat and hesitate are all indicative of the mental processes involved. That is to say, an analysis of the phonological aspects of formulaic sequences may be a window for us to see the processing of formulaic sequences from a different perspective.

This paper provides an overview of the relevant research on the phonological features of formulaic sequences. First, we look at what is said in the literature about the phonological features of formulaic sequences in spontaneous speech, which can be summed up as *phonological coherence*. Then, we consider the psycholinguistic underpinnings of these phonological properties. In the next section, we discuss the implications of phonological coherence for different areas of research, and finally, we look at some recent empirical studies on the phonology of formulaic sequences in spontaneous adult native speaker speech.

Aspects of the Phonology of Formulaic Sequences

While it is widely accepted that formulaic sequences demonstrate some degree of fixedness in lexical form (Zgusta, 1967; Peters, 1983; Baker & McCarthy, 1988; Hickey, 1993; Weinert, 1995; Wray & Namba, 2003), we can perhaps imagine there is also a certain level of fixedness in the phonological form of formulaic sequences, which can be illustrated using Ashby’s (2006) examples. While it is normal to say (1a) and (2a), the variants (1b) and (2b) destroy the idiomatic meaning of the idioms, or as Ashby says, produce humorous effects.

- (1a) She has eyes in the back of her HEAD.
- (1b) She has eyes in the BACK of her head.
- (2a) They’re ROLLing in money.
- (2b) They’re rolling in MONEY.

This fixedness extends beyond stress placement, which is obvious in these two examples, to intonation and tempo. In fact, discussion on these two

aspects has a longer history than the stress placement aspect because the first mentions of intonation and tempo can be traced back to the 1970s in child language literature (e.g. Bloom, 1973; Scollon, 1974; Dore, 1976; Rodgon, 1976; Peters, 1977, 1983). Peters observes that children are able to produce utterances that are above their current expected level of grammatical competence. According to Peters (1983), these utterances are 'always produced fluently as a unit with an unbroken intonation contour and no hesitations for encoding' (p. 8). This is the first use of the term *phonological coherence* in the literature. There are other descriptions of the phonological characteristics elsewhere, for example, these utterances have 'a "melody" unique enough so that [they] can be recognized even if rather badly mumbled' (Peters, 1977, p. 563), and they are articulated fluently but imprecisely by children (Plunkett, 1990). These phonological features provide clues based on which researchers can distinguish utterances that have been learned as a chunk from those that are produced by saying single words in succession. Bloom (1973) traced the emergence of grammar in her daughter's speech from when the child said her first words at nine months to when she uttered sentences at 22 months. Analysing diary records and video recordings of the child, Bloom observes that words said in succession as single word utterances have 'terminal falling pitch contour, and relatively equal stress, and there was a variable but distinct pause between them so that utterance boundaries were clearly marked' (1973, p. 41). Based on similar observations, Peters (1983) includes phonological coherence as one of her list of six criteria for identifying formulaic sequences in child language.¹ This list of identification criteria is advanced by Hickey (1993) who expands the number of criteria to nine and introduces the distinction between the typical and the necessary criteria. Only two out of the nine criteria are categorized as *necessary* (i.e. the satisfaction of the criteria is essential), and phonological coherence is one of them. This shows the importance of the phonological criterion in detecting formulaicity in child language. In fact, Plunkett's (1990) success in applying his *articulatory/fluency criteria* to identify formulaic expressions in the longitudinal data from two Danish children demonstrates how phonological coherence can be used as the sole identification criterion.

Researchers of formulaic sequences in adult language are quick to spot the potential benefits of the finding that formulaic sequences demonstrate unique phonological properties in child language. This finding on child language, combined with the report of Dechert (1983) and Raupach (1984) that distinctively fluent stretches of speech are found enclosed by pauses and/or falling tones, leads some researchers to speculate on the use of phonological coherence to identify formulaic sequences in *adult native*

speaker's speech. Gradually, we see proposals in the literature like 'If a formulaic string is treated as a single, holistic unit, it ought to be relatively resistant to internal dysfluency and inaccuracy . . . we can make the prediction that there would be far fewer pauses and errors within formulaic strings than between them' (Wray, 2004, p. 260), 'multiword items often form single tone units' (Moon, 1997, p. 44) and 'MWUs may be tested by examining whether or not they are amenable to crossing the boundaries of tone-units. [...] MWUs up to clause-length normally occupy one tone-unit and only one' (Baker & McCarthy, 1988, p. 14). However, while there is empirical evidence to support the use of phonological coherence to identify formulaic sequences in child language, researchers have yet to test if the phonological criteria work in adult native speaker speech, and if they are *essential* in the identification. Before empirical evidence is found for adult native speaker speech, these suggestions remain theoretical.

Whether it is valid to directly apply the phonological criteria in the identification of formulaic sequences in adult spoken language is an issue we will return to later in the chapter. Worth noting is the fact that this borrowing of ideas from child language research might have overlooked the phonological aspects not mentioned in the original child language literature. A closer look at the phonological properties above reveals that they mainly concern two aspects of phonology, that is, intonation (i.e. unbroken contour and single tone units) and tempo (i.e. resistance to internal dysfluency, no hesitations, few pauses), but missing the phonemic aspect and the stress placement of formulaic sequences. However, recent studies by Ashby (2006), Bybee (2000, 2001, 2002) and others fill this research gap.

On the phonemic aspect, empirical studies show that high frequency words and phrases undergo phonetic reduction at a faster rate than low and mid frequency sequences (Bybee, 2002, 2006). These include Hooper's (1976) study on English schwa deletion as in *memory* versus *mammary*, Bybee's studies on American English *t/d* deletion (2000) and Spanish intervocalic *d*-deletion (2001), Bybee and Scheibman's (1999) study on the reduction of *don't* in sequences like *I don't know*, *I don't think*, *I don't have*, and Bush's (2001) study on palatalization in sequences like *did you*, *don't you*, *would you* and *that you*. These studies, therefore, lead researchers to believe that formulaic sequences, which tend to have relatively high frequency, experience phonetic reduction more readily than non-formulaic language, which tend to have relatively low frequency.

On the aspect of stress placement, Ashby (2006) studies the accentual patterns of idioms and suggests that idioms can be put into three classes depending on the type of fixedness with their accentual patterns.

His Case (i) idioms have an accentual pattern that is the same as the least-marked literal version, e.g. *to have a CHIP on one's shoulder* compared with *to have a BEE on one's shoulder*. Case (ii) idioms have an accentual pattern different from the corresponding literal expression, e.g. *POUR down* (idiomatic), *pour DOWN* (literal), *be ROLLing in money* (idiomatic), *be rolling in MONEY* (literal). Case (iii) idioms have a very strict accentual pattern and even tone choice is highly constrained, e.g. *I could eat a \HORSE* (falling tone) instead of *I could eat a \HORSE* (falling-rising tone). After analysing sound-play type of jokes involving idioms such as *My EARS are burning*, Ashby suggests that they work by inviting listeners to look for an interpretation other than the obvious. This is achieved prosodically by imposing a narrow focus distinction on the non-compositional part of the idioms. This finding highlights the part played by narrow focus distinction in the phonological fixedness of formulaic sequences.

Psycholinguistic Underpinnings of Phonological Coherence

So far we have reviewed the literature that discusses the phonological features of formulaic sequences. These phonological features, which are manifest in the intonational, temporal, phonemic, and stress aspects, seem to be particular to formulaic sequences. This leads us to speculate that the phonological form of formulaic sequences may be relatively fixed in the same way as the lexical form may be. To explain the formal fixedness of formulaic sequences, the most prominent and popular theory by far is the theory of holistic storage.

The theory of holistic storage

The idea that formulaic sequences are stored and retrieved as unanalysed wholes in the mental lexicon is very powerful and convincing. It explains the phenomena of multiword utterances in child speech (e.g. Peters, 1983; Plunkett, 1990), fossilized errors (Myles, Mitchell, & Hooper, 1999) and distinctively fluent chunks (Dechert, 1983; Raupach, 1984) in language learners' speech, and 'stereotyped, conventional utterances' (Van Lancker & Canter, 1981, p. 64) in some aphasics patients.

In adult native speaker speech, what lends support for holistic storage and retrieval are semantic non-compositionality (i.e. the meaning of the whole does not equal to the sum of meanings of the component words) in idiomatic expressions (e.g. *spill the beans*, *kick the bucket* and *at the end*

of the day), cranberry collocations (e.g. *to and fro* and *put the kibosh on* from Moon, 1998), and syntactically irregular structures which may be passed down from archaic English (e.g. *here comes X*, *attached please find X* and *believe you me*). The theory of holistic storage is also in line with the assumption that formulaic sequences should form single intonation units, have less internal dysfluencies such as hesitations and pauses, be uttered faster than rule-based language, and require specific accentual patterns or focus distinction. While holistic storage is the most popular theory, it is only one among many. There are alternative theories which explain other aspects of the phonological properties of formulaic sequences better than holistic storage. Therefore, it is important that we recognize that the phonology of formulaic sequences is explained by combining the perspectives of different theories.

The theory of holistic storage is the most widely accepted explanation for the behaviour of formulaic sequences which includes formal fixedness, semantic non-compositionality and situation-dependence (i.e. form-function pairing). As far as the phonological aspect is concerned, this theory makes so much sense especially when we consider the case of formulaic sequences in child language. When a child says an utterance with a syntactic structure clearly beyond the developmental stage of the child, and the utterance demonstrates phonological coherence (including distinctive intonation (Peters, 1977), an unbroken intonation contour (Peters, 1983) and articulatory fluency (Plunkett, 1990)), it is highly probable that the whole sequence has been learned as an unanalysed chunk and stored as a holistic unit in the child's mental lexicon. That is to say the theory of holistic storage explains phonological coherence, but at the same time, the observation of phonological coherence also strengthens the creditability of the theory.

The interdependence that exists between the theory of holistic storage and phonological coherence seems logically circular when we point it out, but this is really the logic underlying many researchers' approach to the problem. Some researchers (e.g. Moon, 1997) predict formulaic sequences are phonological coherent because it is believed that they are stored as holistic units, and there are others (e.g. Peters, 1977; Wray, 2004) who infer the holistic storage of formulaic sequences on the basis of phonological coherence. It is difficult to say which of these two approaches is more justifiable or viable than the other. It is perhaps useful to point out that phonological coherence is more of *a fact* because it is measurable. Holistic storage, however, is more of *a claim* because its existence is inferred based on facts. On this foundation, we need to critically evaluate on a case to case basis if the evidence is strong enough for us to infer holistic storage from

phonological evidence. We have seen that it is reasonable to equate phonologically coherent stretches of talk with formulaic sequences in the case of child language because the young child's speech production appears less complex. In the situation of adult speech, however, to equate phonological coherence with formulaic sequences is debatable because given the maturity of their linguistic system and the capacity of their memory, adults are capable of producing a long stretch of speech within the span of a single phonologically coherent unit. This long stretch of speech can be made up of a chain of one or several formulaic sequences linked together by rule-based language. In other words, a phonologically coherent unit can be equivalent to one or more formulaic sequences combined with other single-word items. Ultimately, we are in a dilemma whether to equate a phonologically coherent unit with a formulaic sequence. The equation works well in child language, maybe also in EFL learners' speech, but it is still an unknown if it works in adult native speaker speech.

Chunking and the theory of holistic processing

This issue leads us to consider the idea of *chunks* and *chunking* in psycholinguistics literature, which provides a sound alternative to the theory of holistic storage. In psycholinguistics literature, a chunk is a unit of language processing. A long sentence has to be processed in chunks because of the limitation in the amount of information our short term memory can handle at one time, and this applies to the speaker who has to plan his speech online as well as to the hearer who decodes the speech simultaneously. According to Chafe (1987), an intonation unit represents 'a single focus of consciousness' (1987, p. 32, see above), which is exactly the meaning of a chunk. This is why a chunk and an intonation unit are believed to be equivalent, and both of them are a representation of a processing unit in the brain.

It is important that we note the distinction between holistic storage and holistic processing. Arguably, a unit of storage is more precise than a unit of processing. A unit of storage may be the same as or smaller than a unit of processing. This idea can be illustrated with the help of a metaphor. We can imagine if a colleague brings in a stack of papers which you have requested. If he delivers the whole stack of papers in one go, the whole stack is a processing unit. In the stack, there are journal articles which have been stapled together but there are also loose single sheets. Each detached item, be it a sheet or a stapled collection of sheets, is a storage unit. When we look at the stack of paper stacked together, we know that there are stapled sheets

and loose sheets, but we cannot tell just by looking at the stack of paper whether a particular sheet of paper has been stapled with other sheets or not. We have to be satisfied with 'a stack' as the most precise unit achievable with the information available. If we focus on holistic processing, we are just concerned with the fact that the stack is delivered in one go. This is the same with the idea of the intonation unit. What an intonation unit reveals is a unit of processing, and the fact that a sequence of words is processed as a unit. It does not reveal a unit of storage (and I argue that in reality no psycholinguistic method can ever do that).

By reverting to holistic processing, we can incorporate the eventuality when in adult native speaker speech an intonation unit or a phonologically coherent unit does not map exactly with a formulaic sequence. That can therefore reconcile the divided acceptability of equating formulaic sequences with phonologically coherent units in child language and in adult native speaker speech, which was discussed earlier. With regard to the level of evidence phonological coherence provides, we can be certain that phonological coherence indicates holistic processing. There is the possibility that a unit of storage may span across a whole unit of processing, but a unit of storage can be smaller than a unit of processing – empirical research is needed to determine the distribution of these two cases (see Lin and Adolphs, 2009).

The frequency-based approach

Apart from holistic storage and holistic processing, the frequency-based approach to language also sheds light on the psycholinguistic underpinnings of the phonological features of formulaic sequences, especially concerning phonological reduction (see Bybee, 2002). The frequency-based approach argues that articulation of speech is made up of neuromotor routines whose process can be smoothened by frequent practice (Bybee, 2002). Therefore, the more a series of routines is practiced, the more fluent the articulation is. Therefore, formulaic sequences, which are supposed to be highly frequent in everyday speech, should demonstrate phonological reduction more readily than non-formulaic language. This explains why researchers expect the articulation of formulaic sequences to have higher articulation/speech rate, and markedly fewer dysfluency phenomena like pauses and errors. In fact, this assumption finds support from Dechert (1983) and Raupach's (1984) observations that second language learners can utter distinctively smooth and fluent stretches that seem to be formulaic amid their other dysfluent productions.

So far we looked at the theory of holistic storage, the concept of chunking and the frequency-based approach to language. We discussed how the phonological features of formulaic sequences can be seen from the perspectives of these three areas. As we can see, some phonological features are explained better by one theory than the other (e.g. the frequency-based approach is particularly suited to account for the phonemic aspect of phonological coherence). For this reason, it is important that we combine the strengths of each theory in order to develop a deeper understanding of the phonology of formulaic sequences.

Implications of the Phonology of Formulaic Sequences

Having looked at the psycholinguistic underpinnings of the phonology of formulaic sequences, now we can consider why it matters whether formulaic sequences are phonologically coherent. There are many ways in which researchers, teachers and learners can benefit from a greater understanding of phonological coherence in formulaic sequences, but our focus here particularly concerns three areas. First, as suggested by Baker and McCarthy (1988), Moon (1997) and Wray (2004), phonological coherence may help us identify formulaic sequences in adult speech. Secondly, EFL learners may be able to improve their perceived speech fluency if they also attend to the intonation patterns specific to formulaic sequences (see Wennerstrom, 2001, 2006). Thirdly, researchers of natural language processing (NLP) may be able to make use of this model of the phonology of formulaic sequences to enhance robotic speech and to make it sound even closer to human spontaneous speech.

Identifying formulaic sequences in spontaneous speech has long been a problem in the field of formulaic language research. While it is easy for researchers to give examples of prototypical formulaic sequences, it is difficult to extract formulaic sequences in a corpus which may have as many as a million words. The most popular corpus method nowadays is to use automatic extraction tools to search for highly recurrent word combinations, or *lexical bundles* (Biber et al., 1999). However, many researchers, such as De Cock (1998), have taken note of the fact that amid the prototypical multiword expressions like *I think*, *in fact* and *and things like that*, sequences like *it was it was*, *to to* and *it you*, which may not appear to be psycholinguistic valid, are also retrieved in the corpus results. There are only very few ways of examining the psycholinguistic validity of formulaic sequences,

including eye-tracking (Underwood et al., 2004), self-paced reading (Schmitt & Underwood, 2004; Conklin & Schmitt, 2007), reaction-time experiments (Jiang & Nekrasova, 2007) and so on. All these methods have to be used in an experimental setting using prompts carefully designed in advance to control for intervening variables, and so far they only focus on written formulaic sequences. That said, if phonological coherence can really help researchers examine the psycholinguistic validity of potential formulaic sequences, it will provide a unique window into the processing of spoken formulaic sequences in spontaneous speech. The way this works is, for instance, if lexical bundles extracted from a spoken corpus demonstrate features like a complete alignment with intonation units, lower tendency to contain dysfluency phenomena or errors, or phonological reduction, we can be more certain that the lexical bundles are formulaic. The phonological criteria have to be applied in conjunction with another criterion such as frequency of occurrence because intonation units cannot be equated with formulaic sequences in adult speech as they can in child speech.

Another area which may benefit from a comprehensive investigation of phonological coherence is second language learning. It has long been suggested in the literature that the use of formulaic language is one of the keys to speech fluency in native speakers and language learners alike. The idea is that while some learners of English have to pause to plan their speech online, native speakers would make use of formulaic sequences as fillers (e.g. *I mean, you know* or *you know what I mean*) to buy time for online planning (Skehan, 1998). This way native speakers can maintain what Fillmore (1979) calls *disc jockey fluency* – the ability to keep the speech flow going. It seems that the focus of this idea is on the production of formulaic sequences and it does not concern how formulaic sequences are uttered. However, it is exactly this aspect of phonological coherence that is also important to speech fluency. To illustrate this point with Ashby's (2006) idiom examples again, if the intended is (3a) but the learners say it as (3b) or (3c), it hardly gives an impression of fluency in Lennon's (2000) broader sense of the term. Wennerstrom (2001, 2006) makes a similar point about the important role of intonation in second language fluency. It is clear that the influence of intonation on perceived fluency does not only affect idioms like (3a) but also other forms of conventionalized language or even general spoken English. For instance, if a speaker intends to say 'it rains a LOT in the UK' but actually says 'it rains A lot in the UK', he/she does not come across as a fluent speaker either.² Anyway, this relationship between meaning and accurate corresponding stress placement, which is what

Wennerstrom (2006) called *intonational meaning*, seems to be even more rigid with idioms than with general spoken English, and this point is not immediately obvious to learners of English. Wennerstrom (2006) also suggests that a more holistic style with less self-monitoring and a greater reliance on routinized language chunks would be characterized by longer intonational phrases which would in turn help learners who struggle with word-by-word speech to come across as more fluent speakers. That is why a language syllabus that aims to promote nativelike proficiency and fluency should encourage, not only a greater use of formulaic sequences, but also an awareness of the way formulaic sequences should be said.

(3a) It was raining cats and DOGS.

(3b) It was raining cats AND dogs.

(3c) It was raining CATS and dogs.

The final area which may benefit from a comprehensive study of phonological coherence is natural language processing (NLP) research which models human speech for developing applications in automatized speech synthesis. While the important role of intonational meaning (Wennerstrom, 2006) applies to this area as well, another challenge in this area is to predict the assignment of tempo and pauses, and where an intonation contour begins and ends. In text-to-speech technology, researchers rely mainly on syntactic structure and punctuation to assign temporal and intonation features, but the results are still far from ideal as research by Altenberg (1990a, 1990b) and Knowles and Lawrence (1987) suggest (see below). Chafe's (1988) study also implies that punctuation is at best a rough guide to temporal or intonation units. It does not predict satisfactorily where an intonation unit should begin and end because the length of a punctuation unit (i.e. one punctuation mark every 8.9 words) is almost twice as long as that of an intonation unit (i.e. one intonation unit every 5–6 words).

As various corpus-based research shows, recurrent contiguous word combinations take up 58.6 per cent (Erman & Warren, 2000) to 80 per cent (Altenberg, 1998) of spoken language (depending on the selection criteria, corpus and method of calculation). This high proportion of recurrent sequences, in which formulaic sequences are found, means that if there is a model that describes the phonology of formulaic sequences, a significant proportion of the task to assign phonological features to computer generated language can be tackled.

Over the decades, there have been at least two attempts in the linguistic literature to make computer generated voice more human-like in text-

to-speech synthesis. This is achieved by predicting intonation unit boundaries and tempo based on syntactic parsing. Inspired by Crystal (1975), Altenberg (1990a, 1990b) took the top-down approach and based his prediction of intonation unit boundaries predominantly on grammatical units. This approach has achieved an impressive 93 per cent success rate in predicting boundaries at the between-clauses level but it fails to predict intonation unit boundaries below the between-clauses level including the clause level (i.e. between clause elements) and phrase level (i.e. between phrase constituents). Knowles and Lawrence (1987), however, took the bottom-up approach by starting at the word level and working out when the boundary between each word has to be removed. For instance, the boundary between a 'weak' grammatical word and a lexical word as in *the man* has to be removed. This is the same with combinations like adjective + noun (*old man*) or verb + adverb (*walked slowly*). Like the top-down approach, this bottom-up system of Knowles and Lawrence (1987) also relies on syntactic parsing. That is perhaps because that might well have been the only relatively mature automatic language analysis tool in the late 1980s to early 1990s, and automatic speech synthesis has to be developed on the basis of existing automatic technology available. But the benefits of incorporating insights from formulaic language research are that: (1) formulaic sequences account for at least half of spoken English; (2) formulaic language is a system independent of syntax/grammar so it does not need to be founded on syntactic parsing; and (3) formulaic sequences typically concern the more subtle clause level and phrase level which Altenberg's (1990a, 1990b) project failed to predict. Apparently, future models of speech synthesis in NLP can be enhanced by introducing the concept of formulaic sequences to the current top-down and bottom-up approaches when predicting the assignment of intonation unit boundaries.

Some Recent Empirical Work on the Phonology of Formulaic Sequences in Adult Native Speaker Speech

In the previous sections we have discussed the origin of the idea of phonological coherence and what it entails. In particular reference to phonological coherence in adult native speaker speech, we have examined the different psycholinguistic underpinnings of the phonology of formulaic sequences. This final section will review the few empirical studies in recent years that specifically address the phonology of formulaic sequences in adult native speaker speech.

The Van Lancker, Canter & Terbeek (1981) study

In the literature, Van Lancker, Canter, & Terbeek's (1981) study of the phonological cues which help hearers distinguish accurately whether the speaker intends to convey the idiomatic or literal meaning of the same idioms has often been cited as providing 'the empirical evidence' supporting the use of phonological criteria to identify formulaic sequences in adult spontaneous speech. The five phonological features that Van Lancker et al. (1981) observe have since been taken as a sketch of the phonological profile of formulaic sequences.³ However, this appears to be a misled interpretation of the findings of the study because the Van Lancker et al. made no attempt to claim application of their findings beyond the context they focused on (i.e. reading aloud of idioms in an experimental setting) to cover all types of formulaic sequences in spontaneous speech of adults. In fact, the researchers are also clear about the finding that hearers could not discern with a statistically significant level of accuracy whether the idiomatic or the literal meaning was intended unless the speakers were asked to exaggerate in their own ways the prosodic differences between the two readings. So instead of accepting that the five observed phonological properties (which are the results of the speakers' exaggeration) constitute a simple phonological profile of formulaic sequences, Ashby (2006) offers an alternative interpretation of Van Lancker et al.'s results. His idea, as mentioned earlier, is that there is a certain restriction with introducing focus distinction to the non-compositional part of idioms. If this restriction is tampered with, it serves as a hint to the hearers to look for an interpretation other than the obvious. In other words, what Van Lancker et al.'s (1981) study shows is the different strategies that can be employed by the speakers to tamper with the restriction on focus distinction, and this restriction concerns a small class of semantically non-compositional idioms in the read-aloud experimental condition.

Empirical studies that begin to investigate the broadly defined idioms (i.e. prefabricated language that appear semantically transparent) in spontaneous adult native speaker speech are Dahlmann (2009), Erman (2006, 2007), Lin (2010) and Lin and Adolphs (2009).

The Dahlmann (2009) study

Dahlmann (2009) studies the distribution of pauses around automatically extracted multiword units in her two corpora, the English Native Speaker

Interview Corpus (ENSIC) for adult native speaker speech and the Nottingham International Corpus of Learner English (NICLE) for second/foreign language learner speech. With the support of Raupach (1984) and Wray's (2004) finding, she makes the assumption that pauses should not normally be found within multiword units. This concept then forms the main basis of her phonological criterion when she attempts to develop an inventory of spoken formulaic sequences especially for second/foreign language learners. She extracted lists of multiword units of various lengths automatically from her two corpora and then used the phonological criterion as a filter to select psycholinguistically valid formulaic sequences for inclusion in the inventory.

The Erman (2006, 2007) studies

Erman (2006, 2007) is interested in the distribution of pauses in spontaneous speech collected in the London-Lund Corpus (LLC) and the Bergen Corpus of London Teenager Language (COLT). Her perspective on pauses is that pauses reflect the cognitive processes of speech production and the duration of some types of pauses reflect differences in cognitive processing effort. Therefore, she compares the duration of pauses that come before prefabricated language (or *prefabs*) with those before non-prefabricated, creative language. Her results show that the retrieval of prefabs demands a smaller amount of cognitive processing effort than the retrieval of non-prefabricated language as it is manifest through pause duration. Her study also investigates if there are differences between adolescent and adult speakers in the distribution of pauses over prefabs and non-prefabricated language. Although the results in this respect are not conclusive, there is some indication that the adolescent speakers are more uncertain about their prefabs as indicated in the proportionally higher frequencies of pauses in prefabs.

The Lin and Adolphs (2009) and the Lin (2010) studies

Unlike Dahlmann (2009) and Erman (2006, 2007), a recent study by Lin and Adolphs (2009) is interested in the intonation of formulaic sequences in spontaneous speech of adult *learners* of English. Following what is suggested in the literature (Baker & McCarthy, 1988; Moon, 1997), the study focuses on putting to test the assumption that formulaic

sequences form single intonation units. Using WordSmith 4.0 (Scott, 2004), 56 instances of the five-word lexical bundle *I don't know why* were extracted from the Nottingham International Corpus of Learner English (NICLE). The intonation features of these instances were analysed using both auditory and acoustic approaches. The results of the study reveal that more than half of the instances of *I don't know why* form single intonation units, i.e. they align with the intonation unit boundaries on both sides of the sequence. In 85 per cent of the cases there was matching with the intonation unit boundary on either the left or right side. The significance of their results is two-fold. First, it is the first to test and provide empirical evidence to support the prediction that formulaic sequences often form single intonation units. Secondly, it reveals that there are other many factors that affect whether a formulaic sequence forms a single intonation unit, for instance, the syntactic makeup of the sequence, the position of the sequence in an utterance and the function of the sequences in context. A point to note, however, is that this study investigates the alignment of formulaic sequences with intonation unit boundaries in adult English learner's speech. The results may be different with native speakers.

Lin (2010) extends the investigation of the alignment of formulaic sequences with intonation units in adult native speaker speech. Instead of taking a bird's-eye view of the phonology of a formulaic sequence regardless of the co-text the sequence is in, this study aims at capturing the level of alignment with full attention to the effects of the intervening co-textual and contextual factors such as the neighbouring words, syntactic and lexical structure, the intended emphasis of the utterance, the prosodic features of the speech genre and even the online processing capacity of the speakers. Preliminary results after the analysis of 62 formulaic sequences from a lecture speech extract indicate that 82 per cent of the 62 different formulaic sequences investigated align with intonation unit boundaries on at least one side of the formulaic sequences and 40 per cent align on both sides. Compared with the findings of Lin and Adolphs (2009), these percentages indicate a slightly lower level of alignment of formulaic sequences with intonation units in adult native speaker speech. This lower level of alignment may illustrate the difference in the phonological features of formulaic sequences between native speakers and learners of English, but it can also be put down to differences in genre (Lin and Adolphs, 2009, looks at interview speech but Lin, 2010 examines lecture speech), or other co-textual and contextual factors suggested earlier. For detailed findings and discussion, readers can refer to the study.

Conclusion

In this chapter we reviewed discussions in the literature over the past four decades concerning the phonological features of formulaic sequences, which have been referred to as *phonological coherence* in some literature (e.g. Peters, 1983; Hickey, 1993). We looked at how phonological coherence is manifest in the phonemic, intonation, tempo and stress aspects not only in child language where the notion of phonological coherence originates from, but also in adult native speaker speech. In the earlier sections we also examined the psycholinguistic or cognitive basis of the notion of phonological coherence, and what level of information the phonological evidence provides regarding the holistic storage and processing of formulaic sequences. We considered how researchers of formulaic language, natural language processing, and learners of English can benefit a lot from a deeper understanding of the phonological features of formulaic sequences. Finally, we looked at some recent empirical studies on the phonological coherence demonstrated in formulaic sequences in spontaneous speech of adult native speakers.

In our discussion, we stressed the great need for empirical research on how phonological coherence is manifest in spontaneous adult native speaker speech. While child language researchers began research in this area in 1970s, work that targets adult native speaker speech is still in its infancy. It is expected that a description of the phonological coherence of formulaic sequences in adult speech would be much more difficult because co-textual, contextual, generic and even cognitive factors can also influence the prosodic shape of formulaic sequences in spontaneous speech. That is why a lot of work is waiting to be done on this rich, phonological aspect of formulaic sequences.

Notes

¹ Peter's (1983, pp. 7–12) six criteria for the identification of formulaic sequences in child language are:

- (a) Is the utterance an idiosyncratic chunk that the child uses repeatedly and in exactly the same form?
- (b) Is the construction of the utterance unrelated to any productive pattern in the child's current speech?
- (c) Is the utterance somewhat inappropriate in some of the contexts in which it is used?
- (d) Does the utterance cohere phonologically?
- (e) Is the usage of the expression situationally dependent for the child?
- (f) Is the expression a community-wide formula?

² The author would like to thank Ron Martinez for suggesting the example.

³ Van Lancker et al.'s (1981) 5 observations concerning the phonological features of the idioms include: (1) literal utterances have longer duration than idiomatic utterances (i.e. Idiomatic utterances are spoken faster); (2) literal utterances have 5 times as many pauses compared to idiomatic utterances; (3) juncture (Lehiste, 1973) occurred almost 3 times more in literal than in idiomatic utterances; (4) there are more pitch contours in literal than in idiomatic utterances (0.33 to 0.5 times); and (5) literal utterances are systematically marked by what Bolinger (1965) defined as Accent A.

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